NAME Solution

ChE 210 Quiz #3 6 November 2019

A rigid vessel is initially filled with one mole of a gas mixture of 50 mol% hydrogen (H_2) and 50 mol% acetaldehyde (C_2H_4O) at a total pressure of 760 mmHg absolute and 25˚C. Ethanol (C_2H_6O) forms according to the chemical reaction: $C_2H_4O + H_2 \rightarrow C_2H_6O$

(a) After the system comes to equilibrium at constant temperature of 25˚C, the pressure has dropped to 700 mmHg absolute. Calculate the total moles and the moles of each species present. Assume ideal gas behavior.

K is written wrong!

(b) If the equilibrium expression is given by $K = \left[\frac{y_{C_2H_4O}\, y_{H_2}}{y_{C_2H_6O}}\right]\left(\frac{P_0}{P(atm)}\right)$ where P_0 = 1 atm and $P(atm)$ is the pressure in atmospheres absolute, what is the value of the equilibrium constant K?

(a) $\frac{P_i \cancel{V}}{n_i \cancel{T}} = R = \frac{P_f \cancel{V}}{n_f \cancel{T}}$ $\qquad n_f = n_i\left(\frac{P_f}{P_i}\right) = 1\,mol\left(\frac{700\ mmHg}{760\ mmHg}\right)$

$n_f = 0.921\ mol$

$n_{H_2,f} = n_{H_2,i} - \xi = 0.5 - 0.079 = 0.421\ mol$

$n_{C_2H_4O,f} = n_{C_2H_4O,i} - \xi = 0.5 - 0.079 = 0.421\ mol$

$n_{C_2H_6O,f} = \cancel{n_{C_2H_6O,i}}^{\,0} + \xi = 0.079\ mol$

$n_{tot,f} = n_{tot,i} - \xi$

$\xi = n_{tot,f} - n_{tot,i} = (1 - 0.921)\ mol = 0.079\ mol$

(b) $K = \frac{(n_{C_2H_4O})(n_{H_2})}{(n_{C_2H_6O})(n_{tot})}\left(\frac{1\ atm}{\frac{700}{760}\ atm}\right) = \frac{(0.421)^2}{(0.079)(0.921)}\frac{760}{700} = 2.64$

$= \frac{(0.079)(\cancel{0.921})}{(0.421)^2}\frac{\cancel{760}}{\cancel{700}} = 0.446$

$\cancel{K = \exp(-\Delta G^0/RT) = \exp\left(34{,}750 / 8.3144\ \frac{J}{mol\,K}\ 298K\right)}$

$\cancel{\Delta G^0 = \Delta G^0_{C_2H_6O(g)} - \Delta G^0_{C_2H_4O(g)} - \Delta G^0_{H_2(g)}}$

$\cancel{[(-167.85) - (-133.1) - (0)]\ \frac{kJ}{mol} = -34.75\ \frac{kJ}{mol}}$

$$\text{initial: } V = \frac{n_i R T}{P_i} = \frac{(1\,mol)\left(82{,}057\ \frac{atm\,cm^3}{mol\ K}\right)(298K)}{1\,atm}$$

$$= \underline{\underline{24{,}453\ cm^3}}$$

$$\text{finial: } n_f = \frac{P_f V}{RT} = \frac{\left(\frac{700}{760}atm\right)(24{,}453\ cm^3)}{\left(82{,}057\ \frac{atm\ cm^3}{mol\ K}\right)(298K)}$$

$$n_f = 0.921\ mol\ tot$$

$$V = \frac{n_i R T}{P_i} = \frac{(1\,mol)\left(62.361\ \frac{mmHg\ L}{mol\ K}\right)(298K)}{760\ mmHg}$$

$$= 24{,}452L$$

$$n_f = \frac{P_f V}{RT} = \frac{(700\,mmHg)(24.452L)}{\left(62.361\ \frac{mmHg\ L}{mol\ K}\right)(298K)}$$

$$n_f = 0.921\ mol$$